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ASSIGNMENT
SECTION-A

Ques-1 What is a transaction in DBMS?
Ans-1 A transaction is a logical unit of work consisting of one or more operations performed on a database.
- Includes operations like INSERT, UPDATE, DELECT
- Must be executed completely or not at all
- Ensures data consistency

Ques-2 Explain ACID Properties?
Ans-2 ACID properties ensures reliable transaction processing
- Atomicity → All operations succeed or none
- Consistency → Database remains valid before & after transaction
- Isolation → Transactions do not interfere with each other
- Durability → changes are permanent after commit

Ques-3 What is Concurrency control?
Ans-3 Concurrency control is a mechanism to manage multiple transaction simultaneously without conflicts.
- Prevent data inconsistency
- Ensures correct execution of concurrent transactions.
- Uses techniques like locking, timestamps.

Ques-4 What is a Deadlock?

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Ans-4 A deadlock is a situation where two or more transactions wait indefinitely for each other to release resources.

- No transaction can proceed.
- Occurs due to resource locking.
- Needs detection and recovery.

Ques-5 What is Serializability?

Ans-5 Serializability ensures that the result of concurrent transaction is the same as serial execution.

- Maintains data consistency
- Important in concurrency control
- Equivalent to sequential execution

Ques-6 What is Recovery in DBMS?

Ans-6 Recovery is the process of restoring the database after failure

- Handles system crashes, errors.
- Uses logs and backups
- Ensures no data loss

Ques-7 What is a checkpoint?

Ans-7 A checkpoint is a mechanism that saves the current state of the database.

- Reduces recovery time
- Stores committed transactions
- Avoids reprocessing entire log

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SECTION-B

Ques-1 Explain ACID properties using Banking example.

Ans-1 Scenario - Money transfer from Account A → Account B → (1000)

- Atomicity
  Either both operations happen or none
  - Deduct 1000 Rs from A
  - Add 1000 Rs to B

If one fails → rollback entire transaction

- Consistency
  Total balance remains correct

A+B Total money remains same before and after transcations

- Isolation
  If multiple transfers happen simultaneously

One transcation ~~are~~ should not affect another.

- Durability
  Once transcation is committed

Changes are permanent even after system crash

Ques-2 Illustrate a deadlock situation with two transcation.

Ans-2 Scenario - Two transcations

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- T_1 locks Account A
- T_2 locks Account B

Now,

T_1 wants B

T_2 wants A

Both are waiting forever → Deadlock

Ques-3 Explain serial vs concurrent execution with example.

Ans-3 T_1 : Withdraw Rs 500

T_2 : Deposit Rs 1000

Serial execution

T_1 → completes → then T_2 starts

Example =

Balance = 5000

After T_1 → 4500

After T_2 → 5500

Safe and consistent

Concurrent execution

T_1 and T_2 simultaneously

Possible issues

- Both read same balance (5000).
- Wrong final result may occur

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SECTION-C (Case-based)

Ques-1 A banking system transfers money between accounts. Explain how ACID properties apply.

Ans-1 Scenario - Transfer Rs 1000 from Account A to Account B

Explanation

- Both operations must complete
  - Deduct Rs 1000 from A
  - Add Rs 1000 to B

If one fails → entire transaction is rolled back

- Consistency
  - Total balance remains correct

No money is lost or created

- Isolation
  - If multiple transfers happen at the same time

Each transaction runs independently without affecting others

- Durability
  - Once committed

changes are permanent even after system failure.

Ques-2 Two transactions update same data simultaneously. Identify possible issues.

Scenario -

T_1 - withdraw Rs500

T_2 - Deposit Rs1000

Both access same account balance.

Possible issues

- Lost update Problem

  One transaction overwrites another

final value becomes incorrect

- Dirt Read

  A transcation reads uncommitted data

May lead to wrong decisions

- Non-repeatable Read

  ~~Same~~ Same query gives different results

Data changes during execution

- Occurs due to simultaneous execution without proper control
- solution : locks, isolation levels

SECTION- D (One concept + One Insight)

Ques-1 Explain one ACID property with real-life example.

Ans-1 Atomicity -

Atomicity means a transcation is all or ~~no~~ nothing - either fully completed or fully rolled back.

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Real-life example

Transferring Rs 1000 from Account A → Account B:

- Rs 1000 deducted from A
- Rs 1000 added to B

If system crashes after deduction but before addition:

Transcation is rolled back

Money is restored to A

Insight

Atomicity ensures no partial updates, preventing data inconsistency in real-world systems like banking.

Ques-2 Draw a transcation timeline and identify conflict.

Ans-2 Transcation -

T_1 - Read (A) write (A)

T_2 - Read (A) write (A)

Timeline

Time →

T_1 : Read (A) ..... write (A)

T_2 : Read (A) ..... write (A)

Conflict

Both transcations access same data (A)

Write-write conflicts accurs.

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Problem
Lost update -
Once transcation overwrites the other.
final value becomes incorrect
Insight
Concurrency must be controlled using locking or isolation levels to prevent conflicts.

Ques-3 Explain one recovery technique.
Ans-3 Concept : log-based Recovery
All transcations are recorded in a log file before execution.
How works-
Every operation is stored in log
On system crash
UNDO - incomplete transcations
REDO - committed transcations

T_1 → committed → REDO
T_2 → incomplete → UNDO

Insight
log-based recovery ensures data saftey and consistency, even after system failures.